# Comprehensive Troubleshooting & Technical Report: Pipeline V7 to V9

This document provides a detailed technical breakdown of the critical pipeline failures encountered during the processing of the PRJNA229998 dataset. It documents the exact environmental conflicts, memory constraints, and the iterative shell commands executed to transition the pipeline into a stable, reproducible state.

## Phase 1: Read Desynchronization & Data Extraction

### The Error

The quality control tool `fastp` failed with a fatal error: `ERROR: input files don't contain identical amount of reads` on sample SRR1039513. This was caused by the extraction tool `fasterq-dump` utilizing the `--split-files` flag. SRR1039513 contains over 26.5 million 0-length (empty) reads. The `--split-files` argument drops these empty records asynchronously, resulting in misaligned line counts between the forward (`_1.fastq`) and reverse (`_2.fastq`) files.

### The Rectification Commands

To fix this, the corrupted files were purged, and the reads were re-extracted using the `--split-3` flag, which forces unmated or 0-length reads into a separate singletons file. Line counts were then strictly verified.

```
# 1. Remove the corrupted desynchronized files
rm -f /scratch/osnjila/PRJNA229998_v7_standalone/raw_reads/SRR1039513*.fastq

# 2. Re-extract with --split-3 to guarantee parity
fasterq-dump --split-3 --threads 8 -O raw_reads sra_cache/SRR1039513/
SRR1039513.sra

# 3. Verify exact line count matching
wc -l raw_reads/SRR1039513_1.fastq raw_reads/SRR1039513_2.fastq
# Output confirmed 67,292,352 lines for both files.
```

## Phase 2: Environment & Dependency Management

### The Error

During the transition from SRA extraction to trimming, the pipeline failed with `module not found` and Lmod dependency warnings (e.g., `fastp` complaining about missing `GCC/13.2.0` toolchains). This happens when conflicting C++ libraries are loaded simultaneously or when capitalization of modules on the specific HPC environment is incorrect.

### The Rectification Commands

We implemented strict environment isolation using `module purge` and identified the exact capitalization using the `module avail` command.

```
# Find the exact SRA toolkit module name
module avail sra
# Output revealed "SRA-Toolkit" (capitalized)

# Clean the environment and load the exact compiler toolchain alongside the tools
module purge
module load GCC/13.2.0 fastp STAR Subread
```

## Phase 3: RAM Exhaustion & Interactive Compute Node Allocation

### The Error

During the STAR alignment phase for SRR1039513, the process crashed with `std::bad_alloc`. Loading the GRCh38 STAR index requires ~31.5 GB of memory. Attempting this on the shared interactive login node (`teach-subl`) exceeded memory quotas, resulting in immediate termination by the system.

### The Rectification Commands (Iterative Slurm Allocation)

To safely execute the alignment, we had to request a dedicated interactive compute node. This required iterative refinement of the Slurm `srun` command to satisfy the specific cluster's scheduler constraints:

**Attempt 1: Missing Partition**

```
srun --cpus-per-task=8 --mem=36G --time=01:00:00 --pty bash
srun: error: You must specify a partition (queue) to run in.
```

**Attempt 2: Missing Task Specification**

```
srun -p batch --cpus-per-task=8 --mem=36G --time=01:00:00 --pty bash
srun: error: Unable to allocate resources: Job size specification needs to be
provided
srun: error: You must specify a number of tasks.
```

**Attempt 3: Successful Allocation**

```
srun -p batch -n 1 --cpus-per-task=8 --mem=36G --time=01:00:00 --pty bash
# Prompt successfully changed from osnjila@teach-subl to osnjila@rbl-5
```

### Final Execution on Compute Node (rbl-5)

Once inside the 36GB compute node, the final processing block was executed flawlessly, mapping the reads and chaining directly into the `featureCounts` quantification.

```
cd /scratch/osnjila/PRJNA229998_v7_standalone
module purge
module load GCC/13.2.0 STAR Subread

# Align SRR1039513
STAR --runThreadN 8 --genomeDir ref/star_index   --readFilesIn trimmed_reads/
SRR1039513_1_clean.fastq trimmed_reads/SRR1039513_2_clean.fastq   --
outSAMtype BAM SortedByCoordinate   --outFileNamePrefix alignments/
SRR1039513_

# Quantify full cohort
featureCounts -T 8 -p -B -C -a ref/GRCh38.gtf -o counts/gene_counts.txt
alignments/*.bam
```

## Standard Operating Procedures (SOPs) for Future Runs

- **SRA Extraction standard:** Universally adopt `--split-3` instead of `--split-files`.
- **Job Submission standard:** Never run memory-intensive tasks (like STAR) on a login node. Always use `srun -p batch -n 1 --mem=36G` for interactive testing or `sbatch` for automated execution.
- **Environment Isolation standard:** Prefix all major pipeline phase transitions with `module purge` to flush stale dependencies before loading new toolchains.